Test Series Code

05

All India Test Series
NEET (2025-26)

SET

01

DURATION: 180 Minutes | DROPPER | **M. MARKS: 720**

DATE: 08/02/2026

SYLLABUS

Physics:	Oscillations, Waves, Electric Charges and Fields + 30% Questions from cumulative syllabus of previously conducted all AITS test
Chemistry:	Organic Chemistry: Isomerism, Organic Chemistry: General Organic Chemistry, Hydrocarbon + 30% Questions from cumulative syllabus of previously conducted all AITS test
Botany:	Anatomy of Flowering Plants + 30% Questions from cumulative syllabus of previously conducted all AITS test
Zoology:	Animal Kingdom, Human Reproduction + 30% Questions from cumulative syllabus of previously conducted all AITS test

GENERAL INSTRUCTIONS:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours** duration and the Test Booklet contains **180** multiple choice questions (four options with a single correct answer).
3. There is only one correct response for each question.
4. Each **correct** answer will give **4 Mark** while **1 Mark** will be **deducted** for a **wrong** answer. The maximum marks are 720.
5. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.

OMR INSTRUCTIONS:

1. Use only blue/black dark ballpoint pen.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

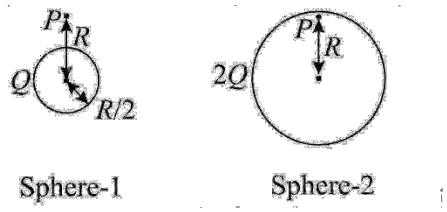
Name of the Student (In CAPITALS): _____

Roll Number: _____

OMR Bar Code Number: _____

Student's Signature: _____ **Invigilator's Signature:** _____

1. Charges Q and $2Q$ are uniformly distributed in two dielectric solid spheres 1 and 2 of radii $\frac{R}{2}$ and R respectively, as shown in the figure. If magnitudes of the electric fields at point P at a distance R from the centre of spheres 1 and 2 are E_1 and E_2 respectively, then:



(1) $E_1 > E_2$ (2) $E_2 > E_1$

(3) $E_1 = E_2$ (4) $E_1 = 2E_2$

2. Five point charges, each of $+10\mu C$, are placed at five vertices of a regular hexagon of side $1\,m$. A point charge of $-1\mu C$ is placed at the center. The net force on the central charge is: (Take $k = 9\times10^9$ SI units)

(1) 0.09 N (2) 0.9 N

(3) 9 N (4) 0 N

3. A travelling wave on a string with linear mass density $\mu = 40$ g/m is given by $y = 0.05\sin(20x - 400t)$ (SI units). The tension in the string is:

(1) 32 N

(2) 20 N

(3) 10 N

(4) 16 N

4. A block of mass 10 kg is hung vertically from a uniform wire having breaking stress of 2×10^6 N/m^2 and density 2×10^4 kg/m^3. What is the maximum length of the wire that can be used to hang the block without the wire breaking under its own weight and the load? (g = 10 m/s^2, Area of cross-section of wire = 3×10^{-4} m^2)

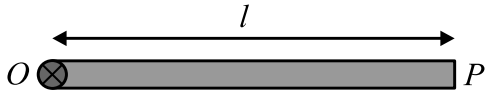
(1) 8.33 m

(2) 4.67 m

(3) 3.67 m

(4) 11.11 m

5. A uniform rod smoothly pivoted at one of its ends is released from rest from horizontal position. If it swings in vertical plane, the maximum speed of the end P of the rod is: (g is acceleration due to gravity)



(1) $2\sqrt{3gl}$ (2) $\sqrt{3gl}$

(3) $2\sqrt{2gl}$ (4) $\sqrt{gl}$

6. A total charge of $10\mu C$ is divided into two parts and placed at a fixed distance. For the electrostatic force between them to be maximum, the two charges must be:

(1) $3\mu C$ and $7\mu C$

(2) $4\mu C$ and $6\mu C$

(3) $1\mu C$ and $9\mu C$

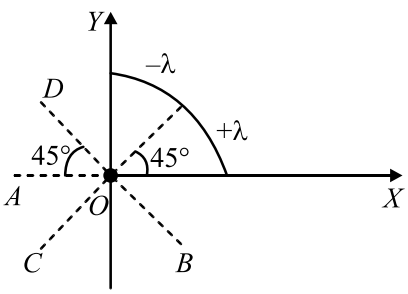
(4) $5\mu C$ and $5\mu C$

7. A particle executes simple harmonic motion between $x = -A$ and $x = +A$. If it takes 2 seconds for the particle to go from the mean position O to $\frac{A}{2}$, how much time will it take to go from $\frac{A}{2}$ to the extreme position A?

(1) 1 s (2) 2 s

(3) 3 s (4) 4 s

8. A charged quarter ring is placed in the X-Y plane with its center at the origin as shown. The ring has uniform positive charge on the half part and the other half has uniformly distributed negative charge. The direction of net electric field due to the quarter ring at the origin is along; (λ is the linear charge density)



(1) OA (2) OB

(3) OC (4) OD

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9. A string clamped at both ends vibrates in its 3rd harmonic. The equation of the wave is $y = A\sin(5\pi x)\cos(100\pi t)$ (SI units). The length of the string is:

(1) 0.6 m

(2) 0.4 m

(3) 1.2 m

(4) 0.2 m

10. A point charge of $\frac{1}{9}\mu C$ is placed at the origin. The electric field intensity at point $(1\text{ m}, \sqrt{3}\text{ m}, 0)$ is:

(1) $125\left(\hat{i} + \sqrt{3}\hat{j}\right)$ N/C

(2) $250\left(\hat{i} + \sqrt{3}\hat{j}\right)$ N/C

(3) $125\left(\sqrt{3}\hat{i} + \hat{j}\right)$ N/C

(4) $500\left(\hat{i} + \hat{j}\right)$ N/C

11. Given below are two statements:

Statement I: Heat absorbed in a cyclic thermodynamic process is always zero.

Statement II: Work done by gas in a cyclic thermodynamics process is always zero.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

(1) Statement I and Statement II both are correct.

(2) Statement I is correct but Statement II is incorrect.

(3) Statement I is incorrect but Statement II is correct.

(4) Statement I and Statement II both are incorrect.

12. Two bodies of equal mass are suspended from two separate massless springs with spring constants 100 N/m and 400 N/m respectively. If the two bodies oscillate vertically such that their maximum velocities are equal, what is the ratio of the amplitude of vibration of the first body to that of the second?

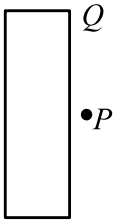
(1) 1 : 2

(2) 2 : 1

(3) 1 : 4

(4) 4 : 1

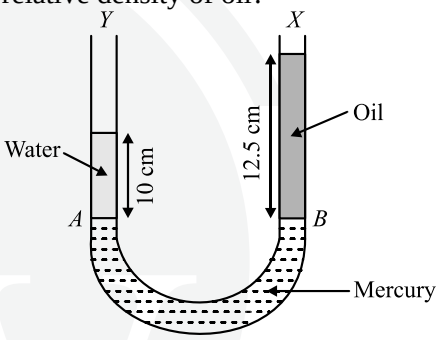
13. A charge Q is given to a large neutral conducting plate of area A as shown below. The net electric field at point P which is located outside near the middle of the plate is:



(1) $\frac{Q}{A\varepsilon_0}$ (2) $\frac{Q}{2A\varepsilon_0}$

(3) $\frac{Q}{4A\varepsilon_0}$ (4) $\frac{2Q}{A\varepsilon_0}$

14. A U-tube contains water and oil separated by mercury. The mercury columns in the two arms are at the same level with 10 cm of water in one arm and 12.5 cm of oil in the other as shown. What is the relative density of oil?

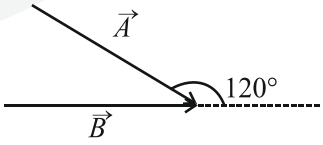


(1) 0.8 (2) 1.2

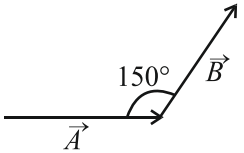
(3) 0.9 (4) 0.6

15. If θ = 30° is the angle between two vectors $\vec{A}$ and $\vec{B}$. The **correct** representation of two vectors making an angle of 30° in a plane is;

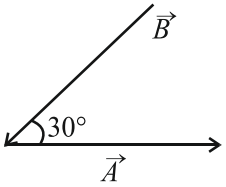
(1)



(2)



(3)



(4) Both (2) and (3)

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16. A progressive wave of frequency 500 Hz is travelling with a velocity of 375 m/s. How far apart are two points 60° out of phase?

(1) 0.250 m
(2) 0.125 m
(3) 0.450 m
(4) 0.200 m

17. Consider an electric field $\vec{E} = (3\times10^3)\hat{i}$ (N/C). What is the flux through a square of side 10 cm, if the normal to its plane makes 60° angle with the positive *x*-axis?

(1) 15 Nm^2/C
(2) 30 Nm^2/C
(3) 10 Nm^2/C
(4) 20 Nm^2/C

18. Given below are the two statements:

Statement-I: As we move closer to a point charge value of electric field may increase if the charge is positive.

Statement-II: An electric dipole is placed in a non-uniform electric field. The net electric force on the dipole will not be zero.

Choose the **correct** answer from the options given below:

(1) Both Statement-I and Statement-II are true
(2) Both Statement-I and Statement-II are false
(3) Statement-I is true but Statement-II is false
(4) Statement-I is false but Statement-II is True

19. A wave travelling in negative *x*-direction in the xy plane with amplitude 0.1 m has a velocity of 360 m/s. If wavelength of wave is 60 m, then **correct** equation for the wave is:

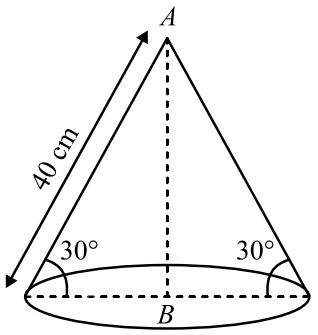
(1) $y = 0.1\sin\left[2\pi\left(6t + \frac{x}{60}\right)\right]$ m

(2) $y = 0.1\sin\left[\pi\left(6t + \frac{x}{60}\right)\right]$ m

(3) $y = 0.1\sin\left[2\pi\left(6t - \frac{x}{60}\right)\right]$ m

(4) $y = 0.1\sin\left[\pi\left(6t - \frac{x}{60}\right)\right]$ m

20. A uniform solid cone with slant height 40 cm is shown in figure. The distance of center of mass of the cone from the center point *B* of the base is:



(1) 10 cm (2) 5 cm
(3) 20 cm (4) 15 cm

21. Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R:

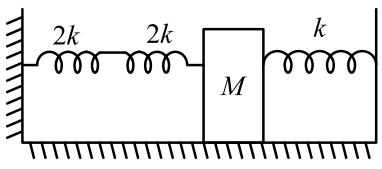
Assertion A: An electric field can never exist inside a hollow conductor but is uniform (non-zero) inside a solid conductor.

Reason R: For a conductor in electrostatic condition electric field inside is zero.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false
(2) A is false but R is true
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is NOT the correct explanation of A.

22. Three massless springs whose force constants are 2*k*, 2*k* and *k* respectively are attached to a mass *M* that is kept on a frictionless plane as shown in figure. If the mass *M* is slightly displaced in the horizontal direction and released, then the frequency of oscillation of system will be:

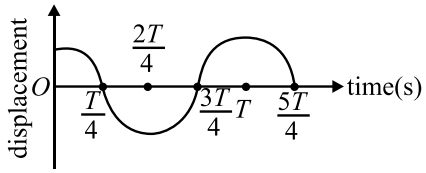


(1) $\frac{1}{2\pi}\sqrt{\frac{k}{M}}$ (2) $\frac{1}{2\pi}\sqrt{\frac{3k}{M}}$

(3) $\frac{1}{2\pi}\sqrt{\frac{4k}{k}}$ (4) $\frac{1}{2\pi}\sqrt{\frac{2k}{M}}$

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23. The displacement time graph of a particle executing S.H.M is given in figure (sketch is schematic and not to scale)



Which of the following statements is/are **true** for this motion?

A. The force is zero at $t = \frac{3T}{4}$

B. The acceleration is maximum at $t = T$

C. The speed is maximum at $t = \frac{T}{4}$

D. The P.E. is equal to K.E. of the oscillation at $t = \frac{T}{2}$

(1) B, C and D
(2) A and D
(3) A, B and C
(4) A, B and D

24. If the differential equation of a particle performing SHM along *X*-axis is given by $\frac{d^2x}{2dt^2} + 16x = 0$, then the time period of SHM is: (here *x* is in meter and *t* is in second)

(1) $\frac{\pi}{2}$ s
(2) $\frac{\pi}{4}$ s
(3) $\sqrt{2}\pi$ s
(4) $\frac{\pi}{2\sqrt{2}}$ s

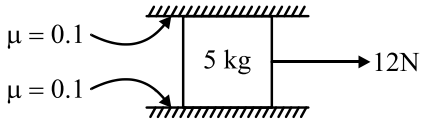
25. The dimensional formula of angular impulse is:

(1) $[ML^{-2}T^{-1}]$ (2) $[ML^2T^{-2}]$
(3) $[MLT^{-1}]$ (4) $[ML^2T^{-1}]$

26. The ratio of average translational kinetic energy to rotational kinetic energy of a linear triatomic gas molecule at temperature *T* is:

(1) $\frac{3}{1}$ (2) $\frac{7}{5}$
(3) $\frac{5}{3}$ (4) $\frac{3}{2}$

27. A block is pulled between the horizontal rough surfaces with horizontal force of 12 N. The acceleration of the block is; (g = 10 m/s^2, Assume the plates creating the rough surfaces are massless)



(1) $\frac{7}{5}$ m/s^2 (2) $\frac{2}{5}$ m/s^2
(3) $\frac{12}{5}$ m/s^2 (4) $\frac{5}{12}$ m/s^2

28. Ratio of amplitudes of two waves of same frequency is 3 : 4. The ratio of maximum and minimum intensity obtained when two waves are superimposed is;

(1) 7 : 1 (2) 49 : 1
(3) 1 : 25 (4) 5 : 1

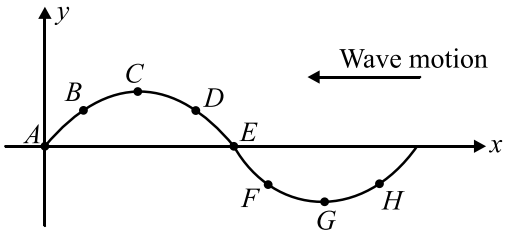
29. For a transverse wave described by $y = A\sin(kx - \omega t)$, the maximum particle velocity is 3 times the wave velocity. The wavelength λ is equal to:

(1) $\frac{2\pi A}{3}$ (2) πA
(3) $3\pi A$ (4) $\frac{\pi A}{3}$

30. A string of mass 0.5 kg and linear mass density 0.1 kg/m is fixed at both ends. If it vibrates at a fundamental frequency of 100 Hz, the wave speed in the string is:

(1) 250 m/s (2) 500 m/s
(3) 100 m/s (4) 1000 m/s

31. A transverse wave is travelling along a stretched string from right to left. The figure shown represents the shape of the string (snapshot) at a given instant. At this instant;



(1) the particles at *A*, *B* and *H* have downward velocity
(2) the particles at *D*, *E* and *F* have downward velocity
(3) the particles at *C*, *E* and *F* have zero velocity
(4) the particles at *A* and *F* have maximum velocity

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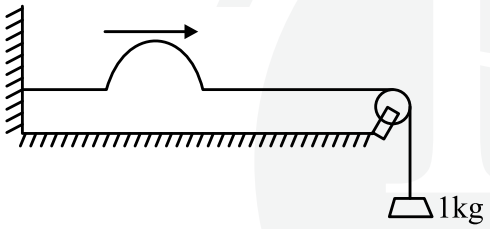
32. The displacement equation of a particle executing simple harmonic motion along x-axis is $x = (6\sin 4t + 8\cos 4t)$. The maximum velocity of the particle will be:

(1) 20 units
(2) 80 units
(3) 40 units
(4) 10 units

33. The displacement x (in m) and time t (in sec) for a particle are related to each other as $t = \sqrt{x} + 3$. What is work done in first six seconds of its motion:

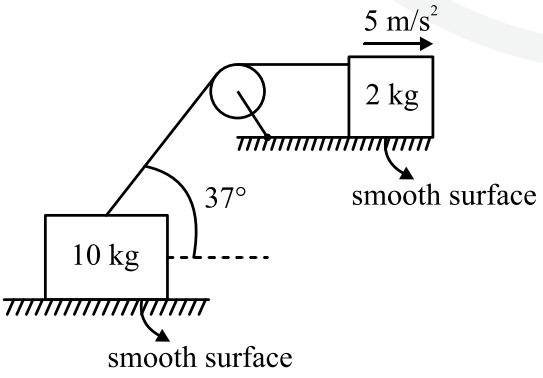
(1) 6 J (2) zero
(3) 4 J (4) 2 J

34. Figure shows a string of linear mass density 1.0 g cm^{-1} on which a wave pulse is travelling. The time taken by the pulse in travelling through a distance of 50 cm on the string is: (Take g = 10 ms^{-2})



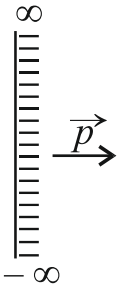
(1) 0.01 s (2) 0.02 s
(3) 0.05 s (4) 0.03 s

35. If the acceleration of 2 kg block at the given instant is 5 m/s^2 then what is the acceleration of the 10 kg block at the same instant? (the string is ideal)



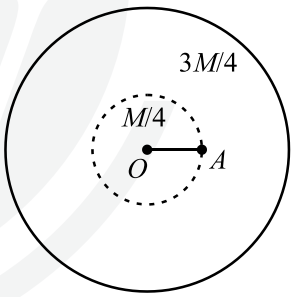
(1) 4 m/s^2
(2) 3 m/s^2
(3) $\frac{25}{3}$ m/s^2
(4) $\frac{25}{4}$ m/s^2

36. An electric dipole is placed perpendicular to an infinite line charge carrying negative charges as shown in figure. Identify the **correct** statement.



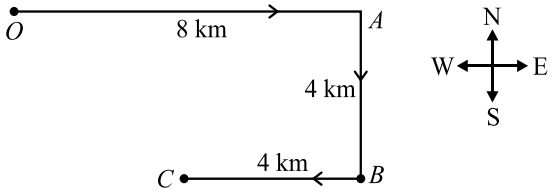
(1) The dipole is attracted towards the line charge.
(2) The dipole is repelled away from the line charge.
(3) The dipole does not experience a force.
(4) The dipole experiences a force as well as a torque.

37. A uniform solid sphere with centre O has mass M. A concentric imaginary hollow sphere is drawn such that it divides solid sphere into two masses $\frac{M}{4}$ and $\frac{3M}{4}$ as shown. Another particle having some mass is placed at point A would experience



(1) a force towards the centre of the solid sphere
(2) a force away from the centre of the solid sphere
(3) no force
(4) the direction of force cannot be determined.

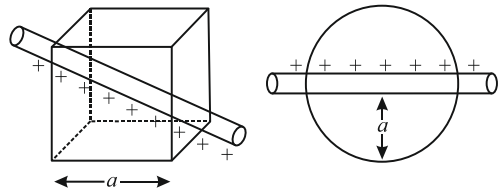
38. A car moves from O to C along the path $OABC$ as shown in figure. The net displacement of the car is:



(1) 4 km, 45° south of east
(2) $4\sqrt{2}$ km, 45° south of east
(3) $4\sqrt{2}$ km 45° north of east
(4) 4 km, 45° north of east

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39. A line charge with linear density λ penetrates a cube of side a diagonally and a sphere of radius a diametrically as shown below. If the electric flux coming out of the sphere is ϕ_0, then the flux coming out of the cube is:



(1) ϕ_0 (2) $2\phi_0$

(3) $\frac{2}{\sqrt{3}}\phi_0$ (4) $\frac{\sqrt{3}}{2}\phi_0$

40. A simple pendulum of length l is suspended from the roof of a trolley which moves in a horizontal direction with acceleration a, then the time period of oscillation is given by $T = 2\pi\sqrt{\frac{l}{P}}$, where P is equal to: (g is acceleration due to gravity)

(1) $\sqrt{g^2 + a^2}$ (2) $g + a$

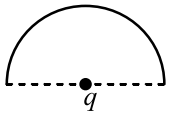
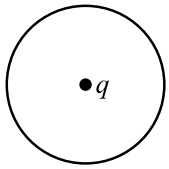
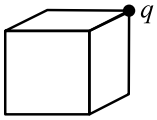
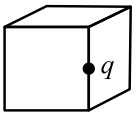
(3) $\sqrt{g^2 - a^2}$ (4) $g - a$

41. The potential energy of a particle of mass 2 kg, moving along the x-axis, is given by the function $U(x) = 10(1 - \cos 2x)$ J. The time period of small oscillations of the particle is:

(1) $\frac{\pi}{\sqrt{5}}s$ (2) $2\pi\sqrt{5}s$

(3) $\frac{\pi}{2\sqrt{5}}s$ (4) $\frac{2\pi}{\sqrt{5}}s$

42. Match the surface given in **List-I** with the electric flux through the surface given in **List-II**.

List-I		List-II	
(A)	Charge q placed at the centre of hemispherical shell. 	(I)	$\frac{q}{\varepsilon_0}$
(B)	Charge q placed at the center of spherical shell. 	(II)	$\frac{q}{2\varepsilon_0}$
(C)	Charge q placed at the corner of a cube. 	(III)	$\frac{q}{4\varepsilon_0}$
(D)	Charge q placed at the centre of a edge of a cube. 	(IV)	$\frac{q}{8\varepsilon_0}$

Choose the **correct** answer from the options given below:

(1) A-II, B-I, C-III, D-IV

(2) A-II, B-I, C-IV, D-III

(3) A-I, B-II, C-IV, D-III

(4) A-I, B-II, C-III, D-IV

43. An oscillator of mass 2 kg and angular frequency 10 rad/s has a Kinetic Energy of 60 J and Potential Energy of 40 J at a specific instant. The amplitude of oscillation is:

(1) 1 m

(2) 2 m

(3) 0.5 m

(4) 0.1 m

44. Three harmonic waves of equal frequency and intensity I_0 have phase angles ωt, $\left(\omega t + \frac{\pi}{3}\right)$ and $\left(\omega t - \frac{\pi}{3}\right)$ respectively. When superimposed, the resultant intensity is:

(1) $4I_0$ (2) $2I_0$

(3) I_0 (4) $9I_0$

45. An infinite line charge produces an electric field of 1.8×10^5 N/C at a perpendicular distance of 10 cm. The linear charge density is:

(1) 1μC/m

(2) 2μC/m

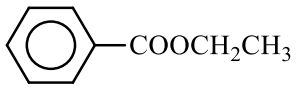
(3) 4μC/m

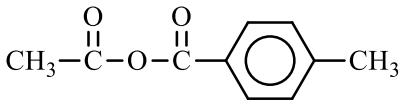
(4) 0.5μC/m

[Contd...

46. Which of the following compounds does **not** show metamerism?

(1) $CH_3CH_2CH_2NHCH_3$

(2) 

(3) 

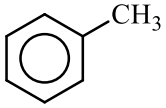
(4) $CH_3CH_2CH_2OH$

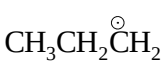
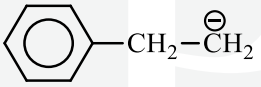
47. Tautomerism is **not** exhibited by:

(1) CH_3CHO (2) $PhCOCH_3$

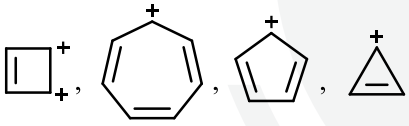
(3) CH_3COCH_3 (4) PhCOPh

48. In which of the following compounds hyperconjugation does **not** take place?

(1)  (2) 

(3)  (4) 

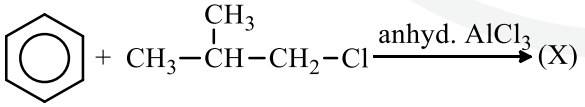
49. How many species among the following are aromatic in nature?



(1) 1 (2) 2

(3) 3 (4) 4

50. Consider the following reaction:



The major product (X) is:

(1) n-butyl benzene (2) tert-butyl benzene

(3) iso-butyl benzene (4) sec-butyl benzene

51. The intermediate involved in the given reaction is:

$CH_2=CH_2 + Br_2 \xrightarrow{CCl_4}$ product

(1) Cyclic bromonium ion

(2) Carbocation

(3) Carbanion

(4) Carbon free radical

52. Oxidation state of phosphorus in $Ca(H_2PO_3)_2$ is:

(1) +1 (2) +3

(3) +1.5 (4) +2.5

53. Given below are two statements:

Statement I: In vacuum all the electromagnetic radiations have different speeds.

Statement II: Black body radiation phenomenon support the wave nature of electromagnetic radiation.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

(1) Both Statement I and Statement II are correct.

(2) Statement I is correct but Statement II is incorrect.

(3) Statement I is incorrect but Statement II is correct.

(4) Both Statement I and Statement II are incorrect.

54. Match **List-I** with **List-II**.

List-I		List-II	
(A)	5.6 L of N_2 at STP	(I)	14 g
(B)	3.022×10^{22} molecule of N_2	(II)	1.4 g
(C)	0.5 g molecule of N_2	(III)	0.28 g
(D)	0.01 mole N_2	(IV)	7 g

Choose the **correct** answer from the options given below:

(1) A-III; B-IV; C-II; D-I

(2) A-IV; B-II; C-I; D-III

(3) A-IV; B-I; C-II; D-III

(4) A-III; B-I; C-IV; D-II

55. In which of the following processes entropy increases?

A. Sublimation of $I_2(s)$

B. Temperature of crystalline solid increased from 30 K to 100 K

C. $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$

Choose the **correct** answer from the options given below:

(1) A and B only (2) B and C only

(3) A and C only (4) A, B and C

56. In a closed vessel containing ammonium bisulphide in equilibrium, ammonia is added such that partial pressure of NH_3 now equals to the original total pressure.

$NH_4HS(s) \rightleftharpoons NH_3(g) + H_2S(g)$

The ratio of the partial pressure of H_2S now to the original partial pressure of H_2S is:

(1) $\frac{1}{3}$ (2) $\frac{1}{2}$

(3) $\frac{1}{4}$ (4) $\frac{1}{6}$

[Contd...

57. IUPAC name of $CH_2=CH-\underset{C_2H_5}{CH}-\underset{CH_3}{CH}-C\equiv CH$ is:

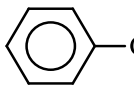
(1) 3-Ethyl-4-methylhex-1-en-5-yne
(2) 4-Ethyl-3-methylhex-5-en-1-yne
(3) 4-Ethyl-3-methylhex-1-yn-5-ene
(4) 3-Ethyl-4-methylhexa-1-ene-5-yne

58. 2.44 g of benzoic acid (molar mass = 122 g mol^{-1}) dissolved in 50 g of benzene (K_f = 5.12 K kg mol^{-1}) shows a depression in freezing point equal to 1.024 K. The degree of dimerisation of benzoic acid in benzene is:

(1) 0.25 (2) 0.50
(3) 1.0 (4) 0.75

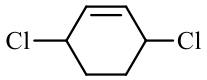
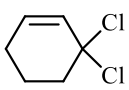
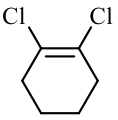
59. Given below are two statements: one is labelled as Assertion A and other is labelled as Reason R:
Assertion A: During electrolysis of aq. NaCl, $Cl_2(g)$ is obtained at anode.
Reason R: On account of overpotential of oxygen, $Cl^-(aq)$ is oxidized to $Cl_2(g)$.
In the light of above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true and R is NOT the correct explanation of A.

60.  and  are related as:

(1) Chain isomers
(2) Position isomers
(3) Functional isomers
(4) Tautomers

61. Geometrical isomerism is shown by:

(1)  (2) 
(3)  (4) 

62. Compound having molecular formula C_3H_9N can show:

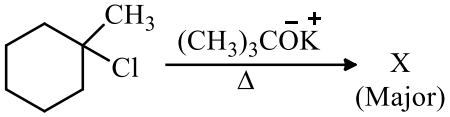
(A) Chain isomerism
(B) Position isomerism
(C) Metamerism isomerism
(D) Functional isomerism

(1) A and B (2) B and C
(3) C and D (4) B and D

63. Maximum number of acidic H-atom are present in:

(1) Ethyne (2) Propyne
(3) But-1-yne (4) But-2-yne

64. Consider the following statements regarding the given reaction:



A It is a β-elimination reaction.
B. X is governed by Sytzeff's rule.
C. Reaction follow E_2 pathway.
The **correct** statements are:

(1) A and B only
(2) B and C only
(3) A and C only
(4) A, B and C

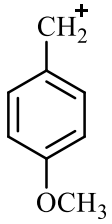
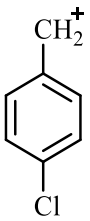
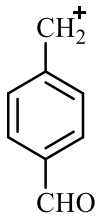
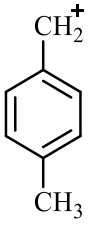
65. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Iso-pentane and neo-pentane	(I)	Tautomers
(B)	Isobutyl chloride and tert-butyl chloride	(II)	Chain isomers
(C)	Phenol and cyclohexa-2,5-dienone	(III)	Functional isomers
(D)	Ethanol and Dimethyl ether	(IV)	Position isomers

Choose the **correct** answer from the options given below:

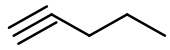
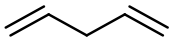
(1) A-III; B-IV; C-II; D-I
(2) A-II; B-IV; C-I; D-III
(3) A-IV; B-I; C-II; D-III
(4) A-III; B-I; C-IV; D-II

66. Most stable carbocation among the following is:

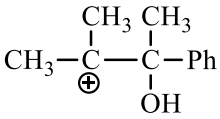
(1)  (2) 
(3)  (4) 

[Contd...

67. Which of the following compounds give 5-keto-3-methylhexanal upon ozonolysis?
(1) 1,2-Dimethylcyclopent-1-ene
(2) 1,3-Dimethylcyclopent-1-ene
(3) 1,4-Dimethylcyclopent-1-ene
(4) 1,5-Dimethylcyclopent-1-ene

68. Which of the following is **not** an isomer of cyclopentene?
(1)  (2) 
(3)  (4) 

69. Which of the following is most likely to migrate to the positively charge carbon in the given carbocation?



(1) H (2) CH_3
(3) Ph (4) No migration

70. Given below are two statements:
Statement I: Iodination of methane is carried out in the presence of oxidizing agent like HIO_3.
Statement II: In the nitration of benzene in presence of HNO_3 and H_2SO_4 mixture, HNO_3 acts as a base.
In the light of the above statements, choose the *most appropriate* answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Both Statement I and Statement II are incorrect.

71. Given below are two statements: one is labelled as Assertion A and other is labelled as Reason R:
Assertion A: Chlorine undergo disproportionation in the alkaline medium.
Reason R: Chlorine is a non-metal.
In the light of above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true and R is NOT the correct explanation of A.

72. Longest wavelength for the transition in the Paschen series of atomic hydrogen is:
(1) $\frac{144}{7R}$ (2) $\frac{36}{5R}$
(3) $\frac{72}{5R}$ (4) $\frac{18}{7R}$

73. For the reaction: $2H_2(g)+O_2(g)\rightarrow 2H_2O(\ell)$ identify the H_2 as a limiting reagent in the following mixtures:
(1) 2 g H_2 + 14 g O_2
(2) 3 g H_2 + 26 g O_2
(3) 4 g H_2 + 30 g O_2
(4) 1 g H_2 + 6 g O_2

74. 3-Ethylpent-2-ene on reaction with HBr in presence of peroxide forms an addition product. The number of possible stereoisomers for the product is:
(1) Zero (2) 2
(3) 4 (4) 6

75. The pH of 0.1 M aqueous $(CH_3COO)_2Ca$ solution will be: (Given: pK_a of CH_3COOH = 4.74)
(1) 8.21 (2) 11.15
(3) 10.63 (4) 9.02

76. Given below are two statements: one is labelled as Assertion A and other is labelled as Reason R:
Assertion A: A mixture of chloroform and acetone forms a solution with negative deviation from Raoult's law.
Reason R: Chloroform molecule is able to form hydrogen bond with acetone molecule.
In the light of above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true and R is NOT the correct explanation of A.

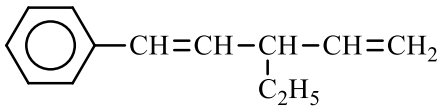
77. The conductivity (κ) of an electrolyte solution depends on:
(A) The concentration of electrolyte
(B) Nature of solvent
(C) Temperature
Choose the **correct** option.
(1) A and B only (2) A and C only
(3) B and C only (4) A, B and C

[Contd...

78. Which of the following compounds can exist as meso form?

(1) Lactic acid (2) Glycine
(3) Tartaric acid (4) Picric acid

79. Total number of stereoisomers of the given compound is:



(1) 2 (2) 4
(3) 6 (4) 8

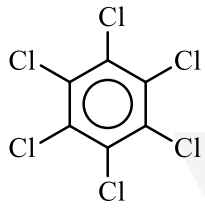
80. Hybridisation of carbon atoms in  are:

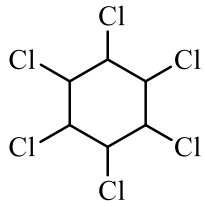
(1) sp^3 and sp^2 (2) sp^3 only
(3) sp only (4) sp^2 only

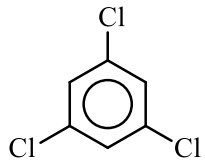
81. In the following compounds C-Cl bond energy is maximum in:

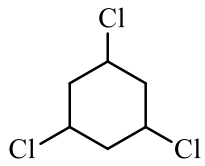
(1) Vinyl chloride (2) Chlorobenzene
(3) Allyl chloride (4) Ethyl chloride

82. Gammaxane is:

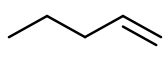
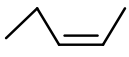
(1) 

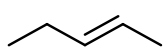
(2) 

(3) 

(4) 

83. Pent-2-yne on partial reduction with Na/liq.NH_3 gives:

(1)  (2) 

(3)  (4) 

84. Given below are two statements:

Statement I: Diastereomers are not related to each other as non-superimposable mirror images to each other.

Statement II: Enantiomers have different physical properties.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

(1) Both Statement I and Statement II are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Both Statement I and Statement II are incorrect.

85. The **correct** decreasing order of –I effect is:

(1) $-F > -CN > -\overset{+}{N}H_3$

(2) $-\overset{+}{N}H_3 > -F > -CN$

(3) $-\overset{+}{N}H_3 > -CN > -F$

(4) $-CN > -F > -\overset{+}{N}H_3$

86. Propyne on warming with $HgSO_4$ and dil.H_2SO_4 at 333 K forms:

(1) Alcohol (2) Aldehyde
(3) Ketone (4) Carboxylic acid

87. In Newman's projection of ethane, the staggered form has dihedral angle of:

(1) 0° (2) 60°
(3) 90° (4) 30°

88. Electrophile and nucleophile respectively are:

(1) H_2O and NH_3 (2) $AlCl_3$ and C_2H_4
(3) Co^{3+} and BF_3 (4) $SnCl_4$ and NO_2

89. Which of the following alkanes has maximum boiling point?

(1) n-Pentane
(2) 2-Methylbutane
(3) 2,2-Dimethylpropane
(4) n-Butane

90. Most stable carbanion among the following is:

(1) $CH_3^{\ominus}$ (2) $CH_3-\overset{\ominus}{C}H_2$
(3) $(CH_3)_2\overset{\ominus}{C}H_2$ (4) $(CH_3)_3C^{\ominus}$

[Contd...

91. Which of the following prizes are part of the triple crown won by Ernst Mayr?
(1) Nobel Prize, Balzan Prize, Crafoord Prize
(2) Balzan Prize, International Prize for Biology, Crafoord Prize
(3) International Prize for Biology, Nobel Prize, Fields Medal
(4) Crafoord Prize, Turing Award, Balzan Prize

92. Consider the following statements regarding parenchyma tissue:
A. It forms the major component within organs.
B. The cells are generally isodiametric.
C. Their walls are thick and made up of lignin.
D. They perform functions like photosynthesis, storage, and secretion.
E. Intercellular spaces are always absent.
The **correct** statements are;
(1) A, B and D
(2) B, C and E
(3) A, C and E
(4) B, D and E

93. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): Apical meristems are a type of primary meristems.
Reason (R): Apical meristems appear early in life of a plant and contribute to the formation of the primary plant body.
In the light of the above statements, choose the *most appropriate* answer from the options given below.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

94. The correct order of taxonomic categories from the lowest to the highest is;
(1) Kingdom → Phylum → Class → Order → Family → Genus → Species
(2) Species → Genus → Family → Order → Class → Phylum → Kingdom
(3) Species → Genus → Order → Family → Class → Phylum → Kingdom
(4) Kingdom → Class → Phylum → Order → Family → Genus → Species

95. Given below are two statements:
Statement I: A mature sieve element possesses a peripheral cytoplasm and a large vacuole but lacks a nucleus.
Statement II: The functions of sieve tubes are controlled by the nucleus of companion cells.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

96. Match **List-I** with **List-II**.

List-I (Cell type)		**List-II (Characteristic)**	
(A)	Tracheids	(I)	Long cylindrical tube-like structure
(B)	Vessels	(II)	Living and thin-walled
(C)	Xylem parenchyma	(III)	Highly thickened walls and obliterated central lumens
(D)	Xylem fibres	(IV)	Elongated or tube like cells with tapering ends

Choose the *most appropriate* answer from the options given below.
(1) A-III, B-IV, C-I, D-II
(2) A-IV, B-II, C-III, D-I
(3) A-I, B-III, C-II, D-IV
(4) A-IV, B-I, C-II, D-III

97. Which of the following is proximal to the region of elongation in a root tip?
(1) Root cap
(2) Region of maturation
(3) Region of meristematic activity
(4) A thimble-like protective structure

98. The chloroplasts are present in:
A. spongy parenchyma cells.
B. guard cells.
C. all sclerenchyma cells.
D. palisade parenchyma cells.
E. some collenchyma cells.
Choose the *most appropriate* answer from the options given below:
(1) A, B, D and E only
(2) B, C, D and E only
(3) A, C and D only
(4) A, B and D only

99. Given below are two statements:
Statement I: Within angiosperms, the monocots and dicots are seen to be anatomically different.
Statement II: Internal structures in plants do not show adaptations to diverse environments.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

100. Who reported that the curly top virus spreads through a plant via the food-conducting tissue?
(1) Antonie Von Leeuwenhoek
(2) Rudolf Virchow
(3) Katherine Esau
(4) George Palade

101. In Whittaker's 5-kingdom classification of organisms;
(1) viruses did not find a place.
(2) kingdom Protista included only unicellular prokaryotes.
(3) members of four kingdoms had organ system level of body organisation.
(4) phylogenetic relationship between them was not considered.

102. Mark the **incorrectly** matched pair.
(1) Fundamental tissue system : Also called ground tissue system
(2) Vascular tissue system : Composed of complex plant tissues
(3) Epidermal tissue system : Forms the outer-most covering of the whole plant body
(4) Conducting tissue system : Composed of all dead cells

103. Match **List-I** with **List-II** w.r.t vascular bundles.

List-I		List-II	
(A)	Radial	(I)	Xylem and phloem jointly situated along same radius
(B)	Conjoint	(II)	Cambium present between phloem and xylem
(C)	Open	(III)	Cambium not present between xylem and phloem
(D)	Closed	(IV)	Xylem and phloem arranged in alternate manner along different radii

Choose the *most appropriate* answer from the options given below.
(1) A-IV, B-I, C-II, D-III
(2) A-IV, B-II, C-III, D-I
(3) A-I, B-III, C-II, D-IV
(4) A-III, B-IV, C-I, D-II

104. The first component of a scientific name;
(1) is the organism's local name.
(2) starts with a capital letter.
(3) is followed by the generic name.
(4) is the name of author who first described that organism.

105. All the following are present in a dicot stem, **except**;
(1) collenchymatous hypodermis.
(2) large pith.
(3) starch sheath.
(4) casparian strips.

106. Match **List-I** with **List-II**.

List-I (Placentation)		List-II (Example)	
(A)	Marginal	(I)	Lemon
(B)	Axile	(II)	Pea
(C)	Parietal	(III)	*Primrose*
(D)	Free central	(IV)	*Argemone*

Choose the *most appropriate* answer from the options given below.
(1) A-II, B-I, C-IV, D-III
(2) A-II, B-III, C-I, D-IV
(3) A-I, B-II, C-III, D-IV
(4) A-IV, B-I, C-II, D-III

107. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): Bulliform cells are empty and colourless.
Reason (R): Bulliform cells are present on adaxial epidermis of grasses.
In the light of the above statements, choose the *most appropriate* answer from the options given below.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

108. Consider the following statements regarding the anatomy of a dicot stem.
A. The vascular bundles are arranged in a ring.
B. Vascular bundles are conjoint and open.
C. Protoxylem is endarch.
D. Endodermis is multilayered.
E. Epidermis is covered with thick layer of cuticle.
The **incorrect** statements are;
(1) A, B, C and E only
(2) B, C and D only
(3) D and E only
(4) C and E only

[Contd...

109. In grasses, the leaves curl inwards to minimise water loss;
(1) due to the activity of lateral meristems.
(2) when bulliform cells become flaccid due to water stress.
(3) due to high pressure in sieve tubes.
(4) because of the presence of many stomata.

110. The lowest common taxonomical category for humans, dogs and cats is;
(1) Phylum Chordata.
(2) Class Mammalia.
(3) Kingdom Animalia.
(4) Order Carnivora.

111. Given below are two statements:
Statement I: Phloem parenchyma is absent in most of the monocotyledons.
Statement II: Phloem fibres (bast fibres) are made up of sclerenchymatous cells.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

112. Choose the **incorrect** statement.
(1) In an isobilateral leaf, stomata are present on both surfaces.
(2) In a dorsiventral leaf, adaxial epidermis generally bears more stomata than the abaxial epidermis.
(3) The ground tissue in both dicot and monocot leaves is called the mesophyll.
(4) The dicot and monocot leaves have some similarities as well as some differences.

113. The cells of sclerenchyma:
A. form the hypodermis of a monocot stem.
B. are usually dead and without protoplasts.
C. are of two types: fibres and sclereids.
D. provide mechanical support to organs.
E. possess pits in cell wall.
How many of the above statements are **incorrect**?
(1) Three (2) Two
(3) One (4) Zero

114. With respect to mode of nutrition, choose the odd one out.
(1) Diatoms (2) *Cycas*
(3) *Agaricus* (4) *Gonyaulax*

115. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): In a dicot root, the pericycle, vascular bundles and pith constitute the stele.
Reason (R): In a dicot root, all tissues on the innerside of the endodermis constitute the stele.
In the light of the above statements, choose the *most appropriate* answer from the options given below.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

116. A plant that shows the presence of secondary vascular tissues;
(1) unexceptionally lacks cork cambium.
(2) will show parallel venation in its leaves.
(3) can have only one cotyledon in its embryo.
(4) is said to have open vascular bundles.

117. Oogamy is observed in;
(1) *Fucus* and *Spirogyra*.
(2) *Volvox* and *Fucus*.
(3) *Volvox* and *Ulothrix*.
(4) *Ulothrix* and *Spirogyra*.

118. Identify the **incorrect** statement regarding stomata.
(1) Stomata regulate the process of transpiration and gaseous exchange.
(2) Each stoma is composed of two guard cells.
(3) In grasses, the guard cells are bean-shaped.
(4) They are a part of epidermal tissue system.

119. Identify the **incorrectly** matched pair.
(1) China rose - Malvaceae
(2) Sunflower - Compositae
(3) Maize - Poaceae
(4) Mustard - Solanaceae

120. Identify the labels that represent strutures that are a part of stomatal apparatus.



(1) A, B and C (2) A, B and D
(3) B, C and D (4) A, C and D

[Contd...

121. Given below are two statements:
Statement I: Strobili are present in *Equisetum*.
Statement II: Heterospory is seen in *Selaginella*.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

122. In a conjoint vascular bundle, phloem is present;
(1) on both sides of xylem.
(2) usually only on the inner side of xylem.
(3) usually only on the outer side of xylem.
(4) surrounded by the xylem.

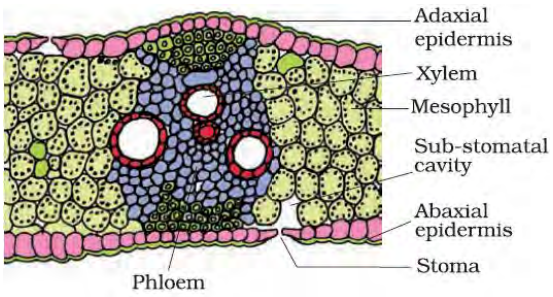
123. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Steroid hormone synthesis	(I)	Smooth endoplasmic reticulum
(B)	Protein synthesis	(II)	Golgi complex
(C)	Glycoprotein synthesis	(III)	Polysomes
(D)	ATP synthesis	(IV)	Mitochondria

Choose the *most appropriate* answer from the options given below.
(1) A-III, B-I, C-II, D-IV
(2) A-III, B-II, C-I, D-IV
(3) A-I, B-III, C-II, D-IV
(4) A-IV, B-I, C-II, D-III

124. Consider the following statements w.r.t permanent tissues in plants.
A. Simple tissues are made of all cells similar in structure and function.
B. Simple tissues are of four types.
C. Complex tissues are of three types.
D. Complex tissues are made of several types of cells.
E. A plant is made of both simple and complex tissues.
The number of **correct** statements is;
(1) five. (2) four.
(3) zero. (4) three.

125. Choose the **correct** statement about the plant part shown below.



(1) It consists of near similar sizes of vascular bundles (except in main veins).
(2) Such a leaf is found in sunflower.
(3) The veinlets form a network.
(4) Such a leaf is called dorsiventral leaf.

126. All the following are lateral meristems, **except**;
(1) Intercalary meristems.
(2) Fascicular vascular cambium.
(3) Interfascicular cambium.
(4) Cork-cambium.

127. A cell having 20 chromosomes undergoes mitosis. Each of its daughter cells will have _____ chromosomes.
(1) 10 (2) 20
(3) 5 (4) 15

128. The radial and tangential walls in cells of casparian strips are deposited with;
(1) lignin. (2) pectin.
(3) suberin. (4) silica.

129. Conjuctive tissue is made of _____ cells.
(1) sclerenchymatous
(2) vascular
(3) collenchymatous
(4) parenchymatous

130. On the stem, the epidermal hairs;
(1) are usually unicellular.
(2) are called trichomes.
(3) are never stiff.
(4) help in increasing water loss through transpiration.

131. Water-containing cavities are present within the vascular bundles of;
(1) dicot stem. (2) monocot stem.
(3) dicot root. (4) monocot root.

132. In dicot root, the endodermal cells are;
(1) bean-shaped. (2) oval or circular.
(3) round. (4) barrel-shaped.

133. Which of the following is not a part of the chromatin?
(1) DNA (2) RNA
(3) Nucleolus (4) Histones

134. The number of xylem bundles in polyarch condition is;
(1) 2-3. (2) 2-4.
(3) 3-5. (4) >6.

135. Meiosis is responsible for;
(1) growth of multicellular organisms.
(2) maintaining the nucleo-cytoplasmic ratio.
(3) replacement of epidermal cells.
(4) bringing about variations.

136. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: In human males, spermatogenesis starts at the age of puberty.

Reason R: There is a significant increase in the secretion of Gonadotropin Releasing Hormone (GnRH) from the hypothalamus at puberty.

In the light of the above statements, choose the **correct** answer from the options given below.

(1) Both A and R are true and R is the correct explanation of A.
(2) Both A and R are true but R is not the correct explanation of A.
(3) A is true but R is false.
(4) A is false but R is true.

137. Read the following statements regarding the phylum coelenterata (Cnidaria):

A. They exhibit tissue level of organization and are diploblastic.
B. Cnidoblasts are used for anchorage, defense, and for the capture of prey.
C. Digestion is extracellular only.
D. They have a central gastro-vascular cavity with a single opening, mouth on hypostome.
E. *Adamsia* (Sea anemone) exhibits the medusa form.

Choose the **correct** option containing only correct statements:

(1) A, B and D only
(2) A, C and E only
(3) B, C and D only
(4) A, B and E only

138. Given below are two statements.

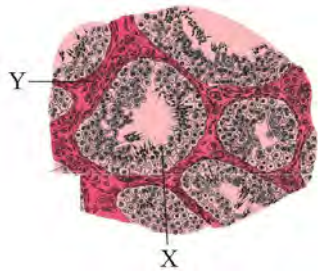
Statement I: The diffusion membrane for gaseous exchange in alveoli is made up of three major layers: the thin squamous epithelium of alveoli, the endothelium of alveolar capillaries, and the basement substance in between them.

Statement II: The total thickness of the diffusion membrane is much less than a millimeter.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

139. Refer to the diagram of the seminiferous tubule given below. Identify the functions of the cells labeled as **'X'** and **'Y'**.



(1) Sertoli cells secrete androgens; Interstitial cells provide nutrition to germ cells.
(2) Sertoli cells undergo meiosis; Interstitial cells secrete FSH.
(3) Sertoli cells provide nutrition to germ cells; Interstitial cells synthesise and secrete androgens.
(4) Sertoli cells secrete LH; Interstitial cells store sperms.

140. Match **List-I** with **List-II**.

List-I		List-II	
(A)	*Balanoglossus*	(I)	Malpighian tubules
(B)	*Pheretima*	(II)	Flame cells
(C)	*Taenia*	(III)	Proboscis gland
(D)	*Locusta*	(IV)	Nephridia

Choose the **correct** answer from the options given below:

(1) A-III, B-IV, C-II, D-I
(2) A-III, B-I, C-II, D-IV
(3) A-IV, B-II, C-I, D-III
(4) A-II, B-IV, C-III, D-I

141. In a standard ECG, the QRS complex represents:

(1) Repolarisation of the ventricles.
(2) Depolarisation of the atria.
(3) Depolarisation of the ventricles.
(4) Repolarisation of the atria.

142. Consider the following statements regarding the menstrual cycle:

A. The menstrual phase lasts for 3-5 days.
B. The follicular phase is characterised by the regeneration of the endometrium through proliferation.
C. LH and FSH attain a peak level in the middle of the cycle (about 14th day).
D. The corpus luteum secretes large amounts of estrogen which is essential for the maintenance of the endometrium.
E. In the absence of fertilisation, the corpus luteum degenerates.

How many of the above statements are **correct**?

(1) Two (2) Three
(3) Four (4) Five

[Contd...

143. Which of the following sets of animals belong to the same phylum?
(1) *Carcharodon*, *Pterophyllum*, *Cucumaria*
(2) *Dentalium*, *Pinctada*, *Chaetopleura*
(3) *Ophiura*, *Antedon*, *Ancylostoma*
(4) *Bombyx*, *Laccifer*, *Nereis*

144. Match **List-I** with **List-II**.

List-I		List-II	
(A)	PCT	(I)	Conditional reabsorption of Na^+ and water
(B)	Descending limb of Henle's Loop	(II)	Reabsorption of 70-80% of electrolytes and water
(C)	Ascending limb of Henle's Loop	(III)	Permeable to water, impermeable to electrolytes
(D)	DCT	(IV)	Impermeable to water, permeable to electrolytes

Choose the **correct** answer from the options given below:
(1) A-II, B-IV, C-III, D-I
(2) A-I, B-III, C-IV, D-II
(3) A-II, B-III, C-IV, D-I
(4) A-II, B-I, C-IV, D-III

145. Given below are two statements.
Statement I: The myometrium of the human uterus exhibits strong contraction during delivery of the baby.
Statement II: The shape of the uterus is like an inverted pear which is supported by tendons attached to the pelvic wall.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

146. Identify the **correct** statements regarding phylum arthropoda:
A. It is the second-largest phylum of Animalia.
B. They possess a chitinous exoskeleton.
C. Circulatory system is of the closed type.
D. Respiratory organs include gills, book gills, book lungs, or tracheal system.
E. Balancing organs called statocysts are present.
(1) A, B and C (2) B, D and E
(3) B, C and E (4) A, D and E

147. During skeletal muscle contraction, which of the following events occurs according to the sliding filament theory?
(1) The 'A' band shortens.
(2) The 'I' band remains the same length.
(3) The 'H' zone disappears or reduces.
(4) The actin filaments slide away from the 'A' band.

148. Select the option where the hormone is **incorrectly** paired with its source in a pregnant female:
(1) hCG – Placenta
(2) hPL – Placenta
(3) Relaxin – Only by placenta (in later phase)
(4) Progesterone – Corpus Luteum (early pregnancy)

149. Which of the following structures is **correctly** associated with the midbrain?
(1) It connects the cerebrum with the spinal cord and contains corpora striata.
(2) It lies between the forebrain and hindbrain and contains the cerebral aqueduct.
(3) It contains the medulla oblongata and regulates respiration.
(4) It forms the major part of the forebrain and encloses the lateral ventricles.

150. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Sponges have a water transport or canal system.
Reason R: The pathway of water transport is helpful in food gathering, respiratory exchange, and removal of waste.
In the light of the above statements, choose the **correct** answer from the options given below.
(1) Both A and R are true and R is the correct explanation of A.
(2) Both A and R are true but R is not the correct explanation of A.
(3) A is true but R is false.
(4) A is false but R is true.

151. Identify the hormone that is a steroid, interacts with intracellular receptors, and regulates gene expression:
(1) Insulin (2) Epinephrine
(3) Cortisol (4) Glucagon

152. Which of the following represents the **correct** sequence of embryonic development stages in humans?
(1) Zygote → Blastocyst → Morula → Gastrula
(2) Zygote → Morula → Blastocyst → Implantation
(3) Zygote → Blastocyst → Gastrula → Morula
(4) Morula → Zygote → Blastocyst → Gastrula

[Contd...

153. Select the **correct** statements regarding the class cyclostomata:
A. They are ectoparasites on some fishes.
B. They have an elongated body bearing 6-15 pairs of gill slits.
C. They possess a sucking and circular mouth with jaws.
D. Their body is devoid of scales and paired fins.
E. They are marine but migrate for spawning to fresh water.
(1) A, B, D and E
(2) A, C, D and E
(3) B, C and D
(4) A, B and C

154. In the male reproductive system of a cockroach, the mushroom gland is present in which abdominal segments?
(1) 4th and 5th
(2) 6th and 7th
(3) 2nd and 3rd
(4) 8th and 9th

155. Which of the following is **not** involved in the foetal ejection reflex?
(1) Fully developed foetus
(2) Placenta
(3) Oxytocin release from maternal pituitary
(4) Prolactin release from maternal pituitary

156. Read the characteristics given below:
A. Skin is dry without glands (except oil gland at tail base).
B. Heart is completely four-chambered.
C. They are warm-blooded (homoiothermous).
D. Air sacs connected to lungs supplement respiration.
E. Bones are hollow with air cavities (pneumatic).
Which vertebrate class is described by the above statements?
(1) Amphibia
(2) Reptilia
(3) Aves
(4) Mammalia

157. If the Tidal Volume is 500 mL, Inspiratory Reserve Volume is 2500 mL, and Expiratory Reserve Volume is 1000 mL, what is the Vital Capacity (VC)?
(1) 3000 mL
(2) 3500 mL
(3) 4000 mL
(4) 1500 mL

158. Given below are two statements.
Statement I: Each spermatogonium is diploid and contains 46 pair of chromosomes.
Statement II: The sex of the baby is determined by the father and not by the mother.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

159. Which of the following animals is an oviparous mammal?
(1) *Macropus*
(2) *Pteropus*
(3) *Ornithorhynchus*
(4) *Delphinus*

160. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: *Erythroblastosis foetalis* can be avoided by administering anti-Rh antibodies to the child immediately after the delivery of the first Rh-positive child.
Reason R: The anti-Rh antibodies destroy the Rh-positive foetal RBCs that might have entered the maternal blood during childbirth, preventing sensitization.
In the light of the above statements, choose the **correct** answer from the options given below.
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is not the correct explanation of A.

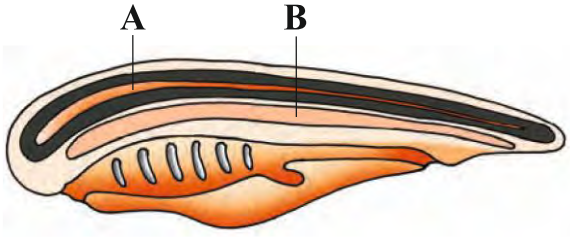
161. During oogenesis, the second meiotic division is completed:
(1) Within the Graafian follicle before ovulation.
(2) Inside the ovary at the time of birth.
(3) During fertilisation, when the sperm comes in contact with the zona pellucida/plasma membrane.
(4) After the formation of the zygote.

162. Given below are two statements.
Statement I: In Urochordata, stomochord is present only in larval tail, while in Cephalochordata, it extends from head to tail region and is persistent throughout their life.
Statement II: *Asterias* (Starfish) is an echinoderm and excretes through a well-developed excretory system.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

163. Which of the following is responsible for the reabsorption of Na^+ and water from the distal parts of the tubule and also leads to an increase in blood pressure?
(1) Atrial Natriuretic Factor (ANF)
(2) Aldosterone
(3) Renin
(4) Angiotensinogen

164. Arrange the following male accessory ducts in the **correct** sequence through which sperm is transported from the testis to the vas deferens:
A. Rete testis
B. Vasa efferentia
C. Epididymis
D. Vas deferens
(1) A → B → C → D
(2) B → A → C → D
(3) A → C → B → D
(4) D → C → B → A

165. Consider the diagram of a Chordate showing fundamental characteristics.



Which of the following **correctly** describes the position of the **'A'** relative to the **'B'**?
(1) The Nerve Cord is ventral and solid, situated below the Notochord.
(2) The Nerve Cord is dorsal and hollow, situated above the Notochord.
(3) The Nerve Cord is dorsal and solid, situated below the Notochord.
(4) The Nerve Cord is ventral and hollow, situated above the Notochord.

166. The joint between the atlas and axis vertebrae which allows the rotation of the head is a type of:
(1) Fibrous joint
(2) Cartilaginous joint
(3) Pivot joint
(4) Saddle joint

167. Which of the following is **not** the function of the Placenta?
(1) Facilitates supply of oxygen and nutrients to the embryo.
(2) Removal of carbon dioxide and excretory material produced by the embryo.
(3) Connects to the embryo through the umbilical cord.
(4) Produces the haploid gametes for the foetus.

168. In which of the following animals are males typically smaller than females (sexual dimorphism) and the alimentary canal is complete with a well-developed muscular pharynx?
(1) *Ascaris*
(2) *Taenia*
(3) *Fasciola*
(4) *Pheretima*

169. Which of the following statements about electrical synapses is **correct**?
(1) Electrical synapses are slower than chemical synapses.
(2) Electrical synapses involve a synaptic cleft where neurotransmitters are released.
(3) Electrical synapses allow direct passage of current from one neuron to another.
(4) Electrical synapses are the most common type of synapse in the human nervous system.

170. Which cell type is **incorrectly** paired with its function?
(1) Stem cells – Potency to give rise to tissues and organs
(2) Inner cell mass – Formation of embryo
(3) Spermatogonia – Undergo meiosis to form sperms
(4) Growing follicles– Secrete progesterone

171. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Addison's disease	(I)	Hypersecretion of growth hormone in adults
(B)	Acromegaly	(II)	Hyposecretion of adrenal cortex hormones
(C)	Cretinism	(III)	Hypothyroidism during pregnancy
(D)	Diabetes mellitus	(IV)	Deficiency of insulin

Choose the **correct** answer from the options given below:

(1) A-I, B-II, C-III, D-IV
(2) A-II, B-III, C-I, D-IV
(3) A-II, B-I, C-III, D-IV
(4) A-IV, B-I, C-III, D-II

172. Read the following statements (A to D) and choose the option which states them as **True** (T) or **False** (F) **correctly**.

A. The endocrine system provides a point-to-point rapid coordination among organs.
B. Molluscs possess longitudinal and circular muscles which help in locomotion.
C. Aquatic annelids like *Nereis* possess lateral appendages, pseudopodia, which help in swimming.
D. In *Scoliodon*, a Chondrichthyes, notochord is persistent throughout life.

	A	**B**	**C**	**D**
(1)	F	F	F	F
(2)	T	T	F	F
(3)	F	F	F	T
(4)	T	F	T	F

173. Regarding the human sperm structure:

A. The sperm head contains an elongated haploid nucleus.
B. The anterior portion of the head is covered by a cap-like structure called acrosome.
C. The middle piece possesses numerous ribosomes to produce energy.
D. The tail facilitates sperm motility essential for fertilisation.

Which statements are **correct**?

(1) A, B and C
(2) A, B and D
(3) B, C and D
(4) A and D only

174. In frogs (*Rana tigrina*), the heart consists of:

(1) Two atria and two ventricles.
(2) One atrium and one ventricle.
(3) Two atria and one ventricle.
(4) One atrium and two ventricles.

175. Match **List-I** with **List-II.**

List-I		List-II	
(A)	*Petromyzon*	(I)	Magur
(B)	*Branchiostoma*	(II)	Lancelet
(C)	*Clarias*	(III)	Lamprey
(D)	*Chelone*	(IV)	Turtle

Choose the **correct** answer from the options given below:

(1) A-III, B-II, C-I, D-IV
(2) A-III, B-II, C-IV, D-I
(3) A-II, B-III, C-I, D-IV
(4) A-I, B-II, C-III, D-IV

176. The partial pressure of oxygen (pO_2) in the alveoli of the lungs is:

(1) Less than that in the oxygenated blood.
(2) Less than that in the deoxygenated blood.
(3) Equal to that in the oxygenated blood.
(4) More than that in the oxygenated blood.

177. Which of the following conditions must be met for fertilisation to occur in humans?

A. The sperm must reach the ampullary region.
B. The ovum must reach the ampullary region.
C. Sperms and ovum must be transported simultaneously.
D. The cervix must be blocked by a mucus plug.

(1) A and B only (2) A, B and C only
(3) C and D only (4) A, B, C and D

178. Given below are two statements.

Statement I: Ammonia is the most toxic form and requires large amount of water for its elimination, whereas uric acid, being the least toxic, can be removed with a minimum loss of water.

Statement II: The property of a living organism to emit light is called bioluminescence and is well-marked in Ctenophores.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

[Contd...

179. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: The Sino-atrial Node (SAN) is called the pacemaker of the heart.

Reason R: The SAN can generate the maximum number of action potentials (70-75 min^{-1}) and is responsible for initiating and maintaining the rhythmic contractile activity of the heart.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both A and R are true and R is the correct explanation of A.

(2) Both A and R are true but R is not the correct explanation of A.

(3) A is true but R is false.

(4) A is false but R is true.

180. The seminal plasma in humans is rich in:

(1) Fructose and calcium but has no enzymes.

(2) Glucose and certain enzymes but has no calcium.

(3) Fructose and certain enzymes but poor in calcium.

(4) Fructose, calcium and certain enzymes.

■■■

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